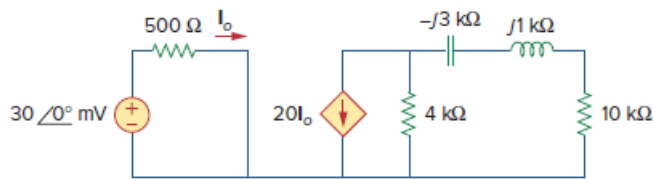


Problem

Obtain the complex power delivered to the 10-k Ω resistor in Fig. 11.77 below.

Figure 11.77:



Step-by-step solution

Step 1 of 6

Refer to Figure 11.77 in the text book.

Calculate the value of the current

$$\begin{aligned} \mathbf{I}_o &= \frac{30\angle 0^\circ \text{ m}}{500} \\ &= 60\angle 0^\circ \mu\text{A} \end{aligned}$$

Therefore, the value of the current $\mathbf{I}_o$ is $60\angle 0^\circ \mu\text{A}$.

Step 2 of 6

Calculate the value of the current passing through 10 k Ω resistor using current division rule.

$$\begin{aligned} \mathbf{I}_{10\text{ k}} &= -20\mathbf{I}_o \left(\frac{4\text{ k}}{4\text{ k} - j3\text{ k} + j1\text{ k} + 10\text{ k}} \right) \\ &= \mathbf{I}_o \left(\frac{-80\text{ k}}{14\text{ k} - j2\text{ k}} \right) \\ &= \mathbf{I}_o (-5.6 - j0.8) \\ &= \mathbf{I}_o (5.657\angle -171.87^\circ) \end{aligned}$$

Substitute $60\angle 0^\circ \mu\text{A}$ for $\mathbf{I}_o$.

$$\begin{aligned} \mathbf{I}_{10\text{ k}} &= (60\angle 0^\circ \mu)(5.657\angle -171.87^\circ) \\ &= (60\angle 0^\circ \mu)(5.657\angle -171.87^\circ) \\ &= 339.42\angle -171.87^\circ \mu\text{A} \end{aligned}$$

Therefore, the value of the current through 10 k Ω resistor is $339.42\angle -171.87^\circ \mu\text{A}$.

Step 3 of 6

Calculate the value of the voltage across $10\text{ k}\Omega$ resistor.

$$V_{10\text{ k}} = I_{10\text{ k}} (10\text{ k})$$

Substitute $339.42\angle -171.87^\circ\text{ }\mu\text{A}$ for $I_{10\text{ k}}$.

$$\begin{aligned} V_{10\text{ k}} &= (339.42\angle -171.87^\circ\text{ }\mu)(10\text{ k}) \\ &= 3.394\angle -171.87^\circ\text{ V} \end{aligned}$$

Therefore, the value of the voltage across $10\text{ k}\Omega$ resistor is $3.394\angle -171.87^\circ\text{ V}$.

Step 4 of 6

Calculate the complex power delivered to $10\text{ k}\Omega$ resistor.

$$S = V_{10\text{ k}} (I_{10\text{ k}})^*$$

Substitute $339.42\angle -171.87^\circ\text{ }\mu\text{A}$ for $I_{10\text{ k}}$ and $3.394\angle -171.87^\circ\text{ V}$ for $V_{10\text{ k}}$.

$$\begin{aligned} S &= (3.394\angle -171.87^\circ)(339.42\angle -171.87^\circ\text{ }\mu)^* \\ &= (3.394\angle -171.87^\circ)(339.42\angle 171.87^\circ\text{ }\mu) \\ &= 1.152\angle 0^\circ\text{ mVA} \\ &= 1.152\text{ mVA} \end{aligned}$$

Therefore, the value of the complex power delivered to $10\text{ k}\Omega$ resistor is 1.152 mVA.

Step 5 of 6

MATLAB calculations:

Write the following instructions in MATLAB m-file and save the program.

format long % to view the exact results

io = (0.03/500)% current io

i10k = -20*io*(4000/(4000-3000i+1000i+10000))% current through 10k ohm resistor

v10k = i10k*10000%voltage across 10k ohm resistor

S10k = v10k*conj(i10k)%complex power delivered to 10k ohm resistor

Step 6 of 6

Run the program and view the results in the command window.

io =

6.000000000000000e-005

i10k =

-3.359999999999999e-004 -4.799999999999999e-005i

v10k =

-3.359999999999999 - 0.480000000000000i

s10k =

0.001152000000000

Therefore, the value of the complex power delivered to $10\text{ k}\Omega$ resistor is 1.152 mVA.